

TUSCOLA AREA AP, MI, USA

WMO: 722007

Lat: 43.459N

Lon: 83.446W

Elev: 701

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54828

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.7	(c) 2.6	(d) -10.7	(e) 3.2	(f) -2.8	(g) -5.1	(h) 4.3	(i) 4.3	(j) 24.3	(k) 23.4	(l) 22.0	(m) 26.8	(n) 3.9	(o) 300	(p) 0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	89.6	72.4	86.3	70.8	83.5	69.9	75.3	83.9	73.6	82.2	72.2	80.1	9.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	124.6	80.2	71.4	119.1	78.0	69.8	112.3	76.7	39.2	84.3	37.6	82.0	36.3	79.8	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 18.6	(c) 16.6	(e) -12.0	(f) 94.0	(g) 7.7	(h) 2.9	(i) -17.6	(j) 96.1	(k) -22.1	(l) 97.8	(m) -26.4	(n) 99.5	(o) -32.1	(p) 101.6		
(4) 20.9	(5) 18.6	(6) 16.6	(7) -12.5	(8) 78.6	(9) 7.5	(10) 1.9	(11) -18.0	(12) 80.0	(13) -22.4	(14) 81.1	(15) -26.6	(16) 82.2	(17) -32.1	(18) 83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	22.8	23.5	33.8	45.2	57.5	66.0	70.3	68.5	61.6	50.5	38.6	29.3		
	DBStd	19.03	11.46	11.49	11.69	8.84	8.87	6.82	5.91	5.29	7.56	8.72	8.99	9.16		
	HDD50	3431	843	741	519	196	26	0	0	0	4	103	357	642		
	HDD65	6919	1307	1161	968	595	268	68	18	26	153	457	792	1106		
	CDD50	2497	0	0	17	52	260	481	628	573	352	119	14	1		
	CDD65	511	0	0	1	1	37	98	181	134	51	8	0	0		
	CDH74	5209	0	0	13	34	462	1061	1890	1206	472	71	0	0		
	CDH80	1506	0	0	2	3	115	326	622	307	120	11	0	0		
(14) Wind	WSAvg	6.6	8.1	8.0	7.6	8.0	6.3	5.4	4.5	4.3	4.7	6.5	7.5	8.0		
(15) Precipitation	PrecAvg	33.1	2.1	1.8	2.0	3.0	3.5	3.6	3.5	2.9	3.0	3.1	2.7	2.1		
	PrecMax	41.4	4.0	4.3	4.1	6.2	7.3	6.3	5.7	4.2	6.1	6.3	5.5	4.2		
	PrecMin	28.3	0.5	0.6	0.6	1.2	1.7	1.3	0.6	1.3	0.5	0.8	0.6	0.6		
	PrecStd	4.0	0.9	1.0	0.9	1.3	1.4	1.4	1.4	0.9	1.6	1.3	1.3	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	57.3	74.7	79.2	87.8	91.1	93.4	90.5	89.6	81.7	67.9	57.7		
		MCWB	50.5	51.3	61.6	59.2	69.6	72.3	75.5	71.9	70.6	70.1	56.6	54.5		
	2%	DB	47.0	49.7	64.1	72.6	83.2	87.9	89.9	86.4	83.3	75.1	61.5	50.3		
		MCWB	43.9	45.8	55.0	55.7	67.2	70.7	73.5	71.4	70.5	63.1	52.7	47.1		
	5%	DB	42.1	42.8	57.2	67.8	79.9	83.7	86.3	83.1	79.4	70.3	57.0	45.5		
		MCWB	39.1	38.8	50.0	54.7	65.4	69.4	71.4	70.5	67.6	62.0	50.4	42.0		
	10%	DB	37.5	37.5	51.7	62.5	75.0	80.5	83.2	80.8	75.7	65.6	53.4	41.1		
		MCWB	35.3	34.3	45.3	51.2	61.5	66.7	70.3	68.7	65.3	59.1	48.1	38.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	53.2	62.2	63.7	72.3	75.3	78.3	77.2	75.9	71.0	59.9	55.0		
		MCDB	52.9	56.1	74.3	73.8	83.5	84.6	88.6	86.1	84.3	79.5	64.5	57.6		
	2%	WB	44.6	45.9	56.9	59.4	69.8	73.2	75.8	74.4	72.5	67.2	54.8	47.8		
		MCDB	46.7	49.4	62.9	67.9	79.2	83.3	85.1	81.7	79.7	72.6	59.6	50.2		
	5%	WB	39.5	39.4	51.4	55.9	67.4	71.4	74.1	72.6	70.0	63.2	51.1	42.6		
		MCDB	42.0	42.4	58.1	64.9	75.9	80.0	82.7	80.0	75.8	67.5	56.1	45.1		
	10%	WB	36.0	35.1	45.2	52.2	64.3	69.4	72.4	70.9	67.8	59.6	47.8	38.1		
		MCDB	37.9	37.8	50.4	61.8	72.6	77.0	80.4	77.7	73.7	64.6	52.8	40.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	15.8	17.7	20.8	22.5	22.7	22.6	21.4	21.9	18.4	15.3	11.4		
		MCDBR	16.6	18.1	26.3	29.6	27.2	25.9	25.4	24.2	25.0	25.1	21.9	16.9		
		MCWBR	14.4	15.1	17.7	16.5	14.2	12.7	11.7	11.7	13.0	14.6	15.5	14.4		
		MCDBR	15.9	17.3	24.4	25.4	24.0	21.7	21.4	20.2	21.2	21.5	19.4	16.6		
	5% WB	MCWBR	14.6	15.0	17.8	16.9	13.9	12.4	11.0	11.1	12.3	14.6	15.5	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.309	0.334	0.367	0.392	0.419	0.410	0.402	0.373	0.347	0.310	0.286		
	taud		2.383	2.328	2.373	2.307	2.292	2.244	2.289	2.300	2.376	2.437	2.496	2.441		
	Ebn at Noon		267	280	287	285	279	271	272	271	270	264	259	259		
	Edh at Noon		24	31	34	39	41	43	41	39	33	28	22	21		
(44) All-Sky Solar Radiation	RadAvg		443	722	1121	1452	1706	1920	1922	1644	1307	821	499	351		
	RadStd		48	96	117	123	145	104	90	67	125	77	63	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (55 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page